Accelerating Fossil Gas Independence in Europe

ECEMP 2026

October 21st 2026

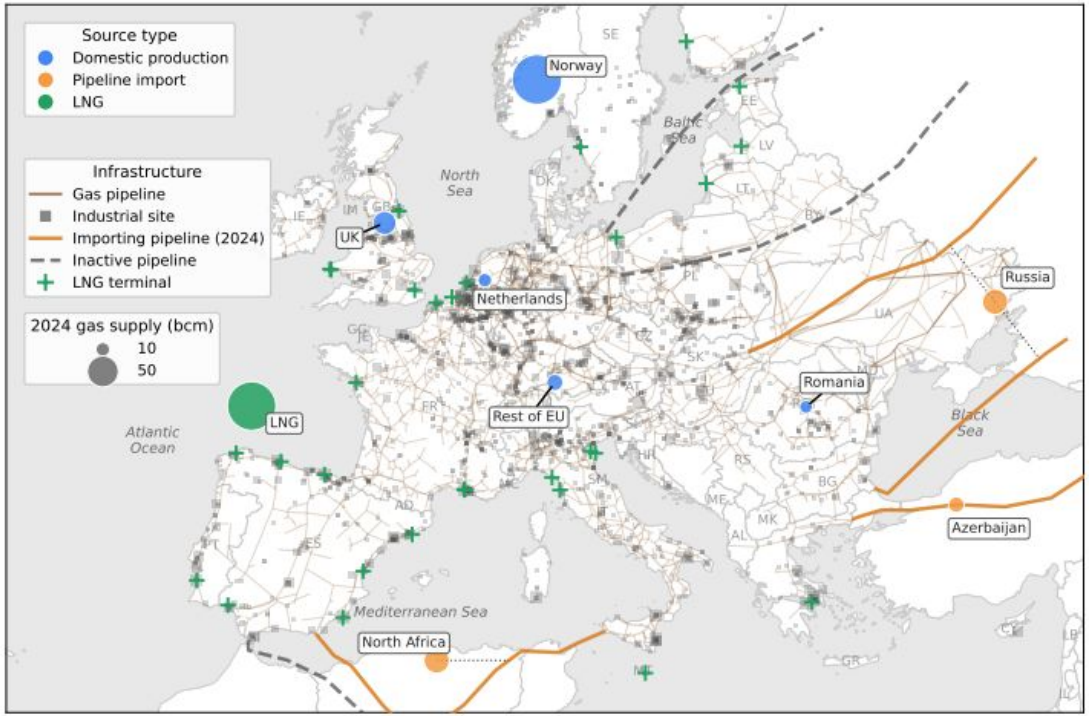
Lukas Franken

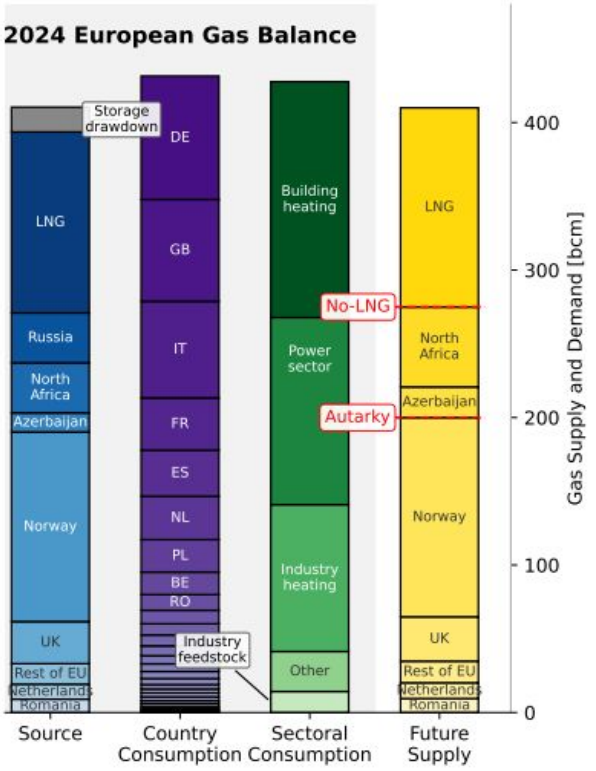
from working paper with
Iegor Riepin and Tom Brown

[Full Paper Link]

European Gas Consumption and Supply

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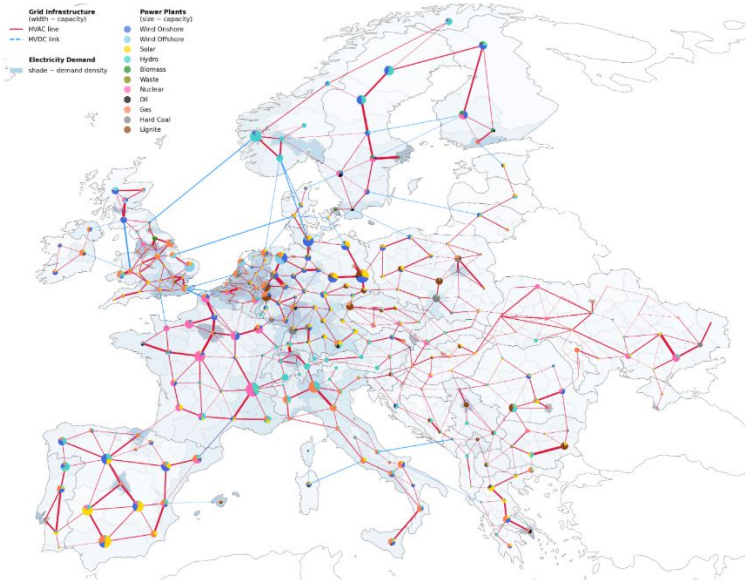


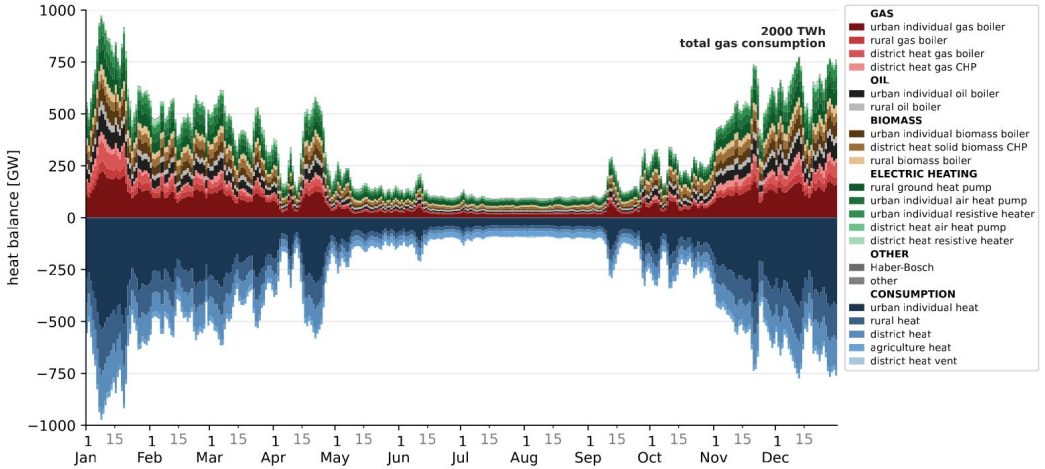


[pre-print]

The PyPSA-Eur model

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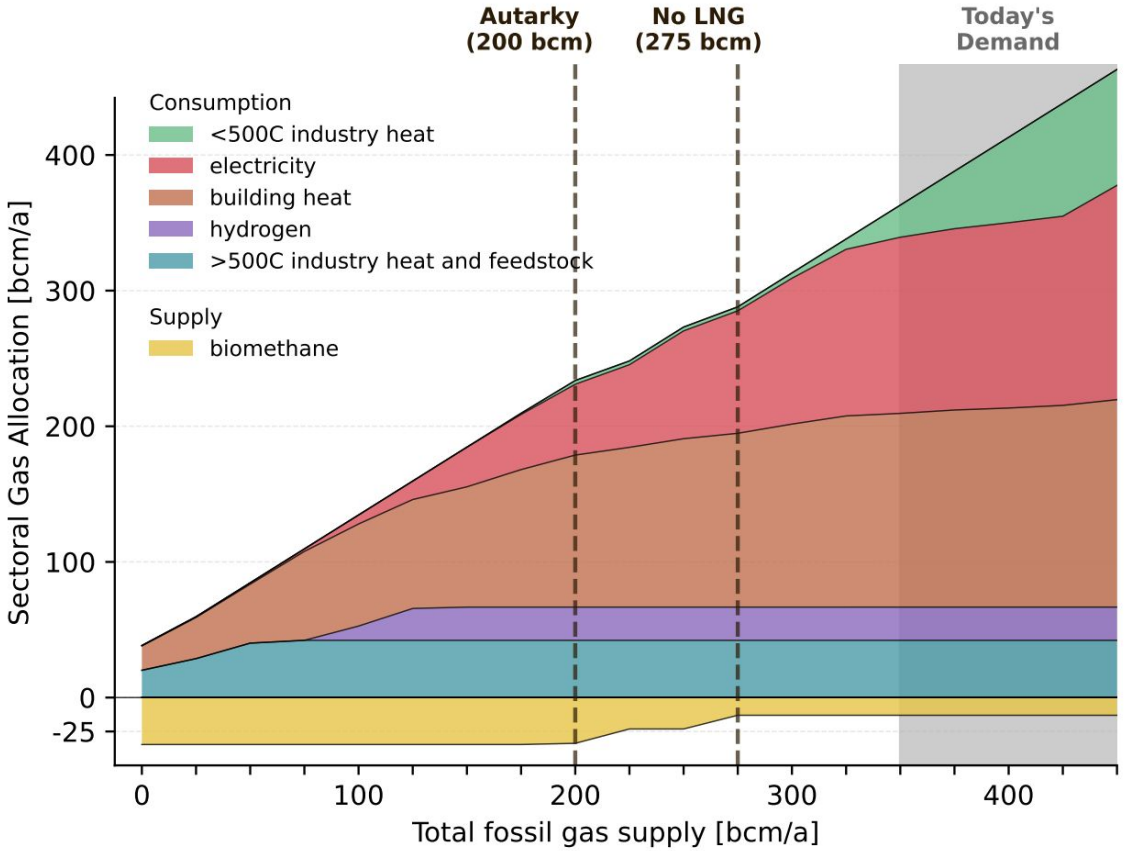




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Constraining Gas Consumption in the Model

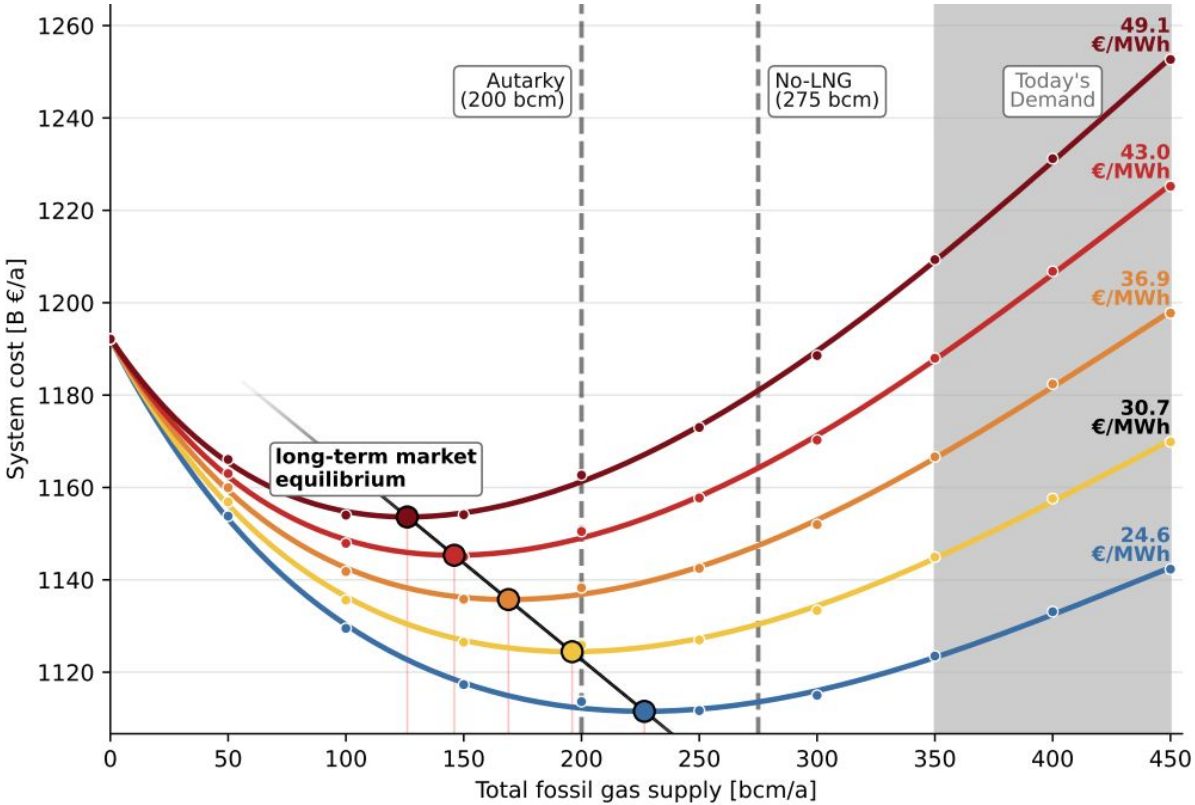
PyPSA Labs



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Gas Abatement Curve

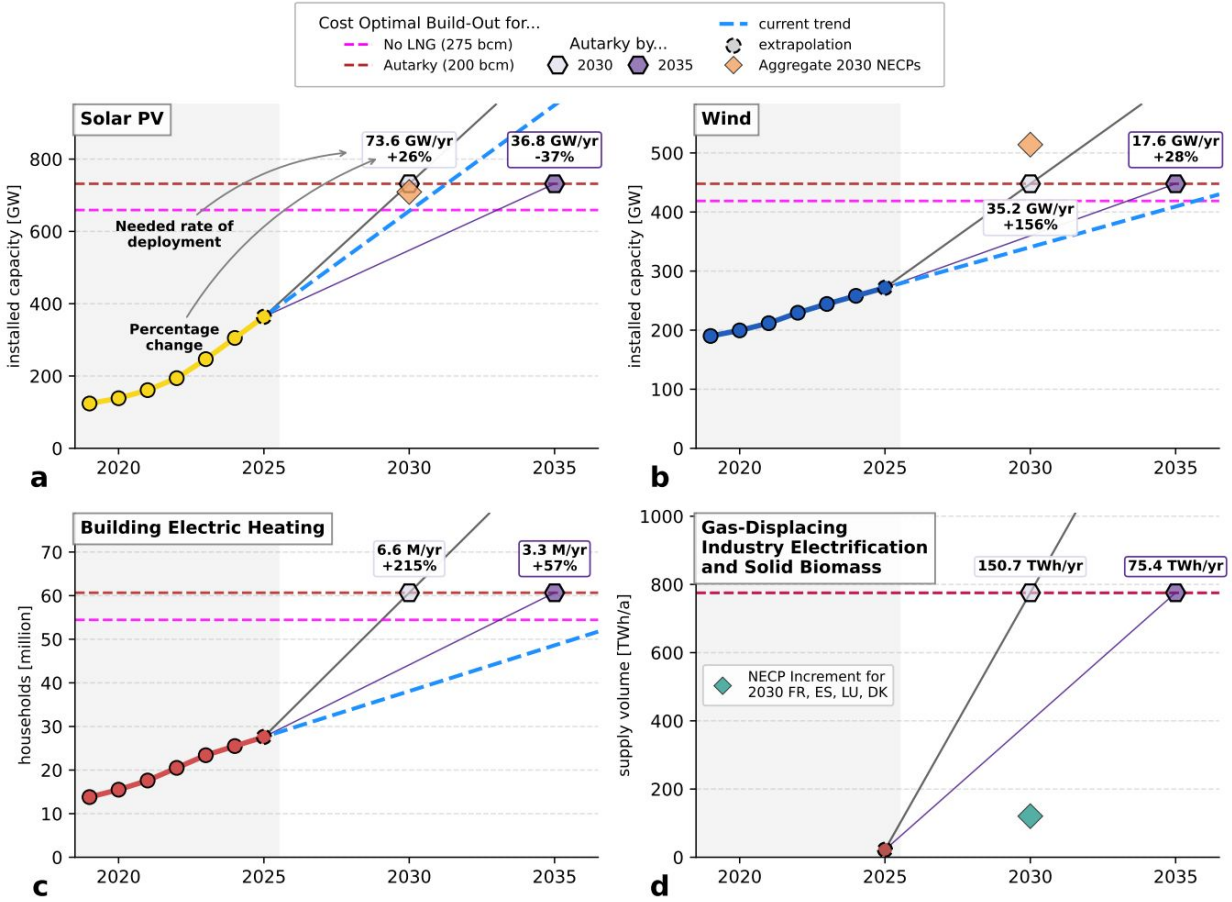
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Current Pace of Roll-out

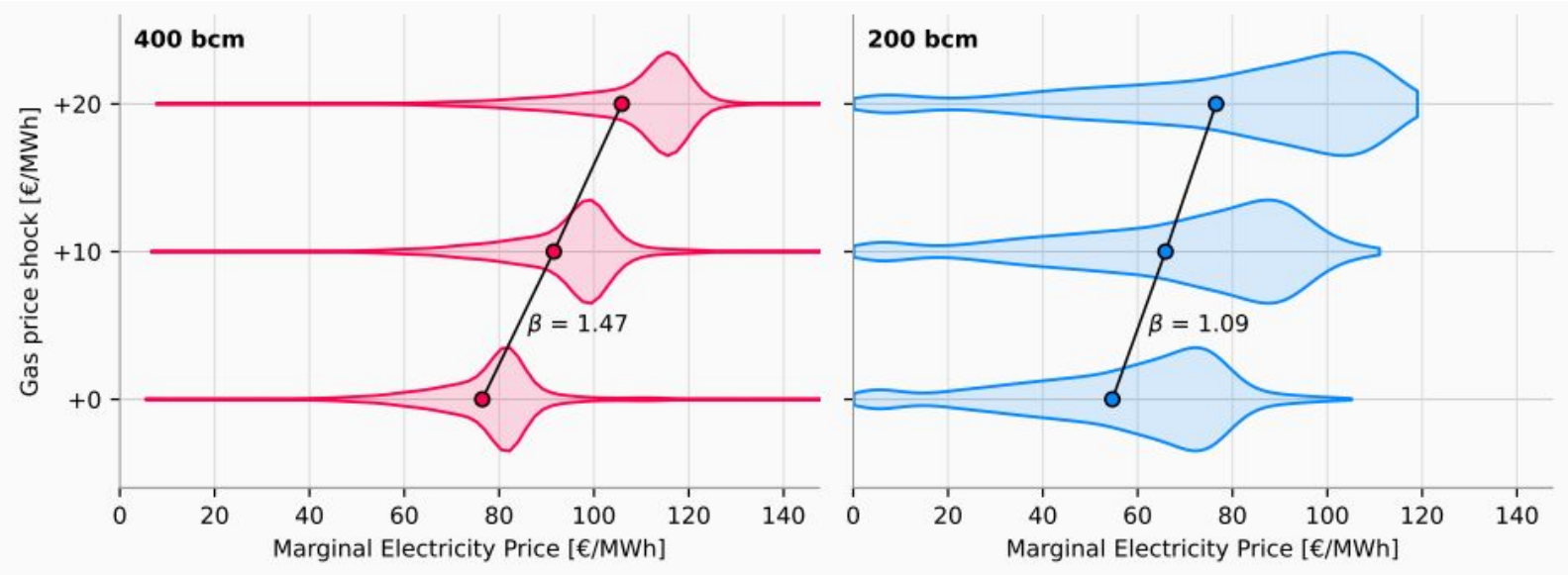
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The Price of Electricity Remains Coupled to the Price of Gas

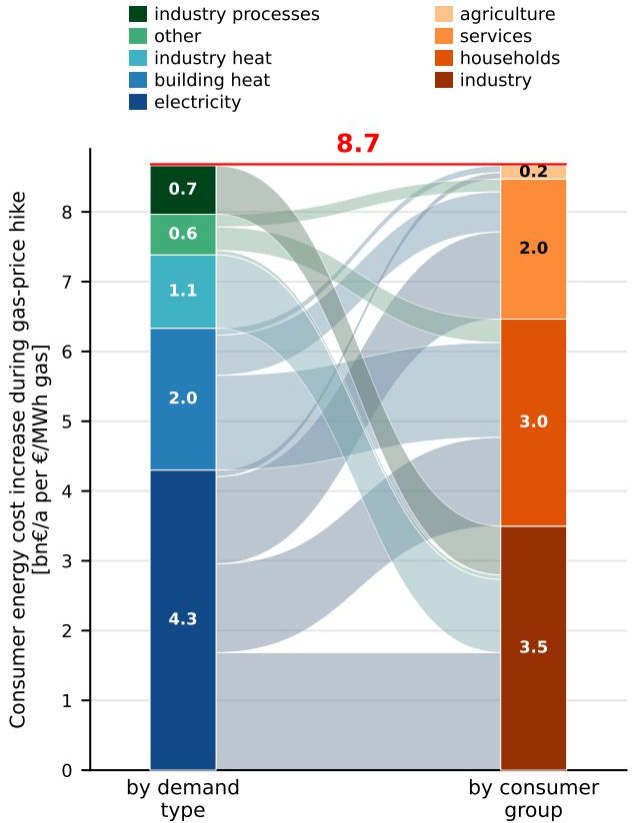
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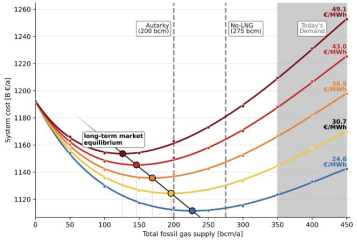
Consumer Impact

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Conclusions



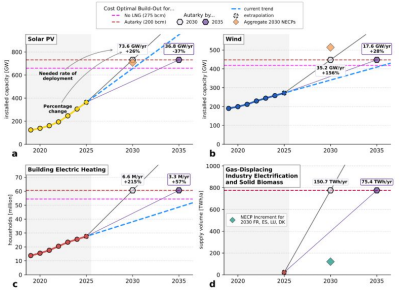
- Reducing gas demand is **cheap, deep into the phase-out.**
- With near-term CO2 pricing, the market **already seeks gas-independence**, but non-economic hurdles slow this transition, at current pace Europe will remain import-dependent **until 2040.**
- Another risk is the **coupling between gas and electricity prices**, which we find to persist deep into phasing out gas, leaving consumers exposed to gas prices even at much lower gas usage.

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Conclusions

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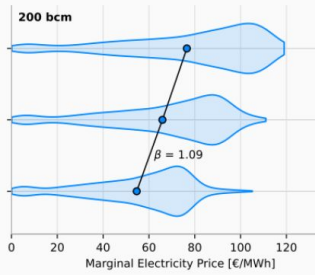


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Conclusions

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Thank you for your attention! Questions?